

Functional Annotation of *opgG* (Q88D03 / PP_5026) in *Pseudomonas putida* KT2440

Target: UniProt Q88D03 · Gene **opgG** (synonym *mgoG*; ordered locus PP_5026) **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), PSEPK **Protein:** Glucans biosynthesis protein G; a secreted precursor of the OpgD/OpgG family

Summary

OpgG (Q88D03; locus PP_5026; synonym *mgoG*) is a periplasmic, Sec-exported carbohydrate-active enzyme of the OpgD/OpgG family that is required for the biosynthesis of osmoregulated periplasmic glucans (OPGs). Together with its operon partner OpgH (the inner-membrane glucosyltransferase PP_5025), OpgG synthesizes and shapes OPGs — anionic β -1,2-linked glucose oligosaccharides bearing β -1,6 branches (and, in pseudomonads, succinyl substituents) that accumulate in the periplasmic space at low medium osmolarity. Recent structural and enzymatic work in *Escherichia coli* has redefined the biochemical identity of this protein: OpgG and its paralog OpgD are **β -1,2-glucanases**, and together they establish a **new glycoside hydrolase family, GH186**. This assigns OpgG a specific catalytic role — hydrolytic/processing action on the β -1,2-glucan backbone — rather than a purely scaffolding function.

The gene identity was rigorously verified. The UniProt annotation (Q88D03, "Glucans biosynthesis protein G," gene *opgG/mgoG*, family OpgD/OpgG) matches the protein's domain architecture and the surrounding genomic context. Global sequence alignment showed that *P. putida* OpgG is **63.6% identical to the crystallized *E. coli* OpgG (P33136)** and **66.4% identical to *P. aeruginosa* OpgG (Q9HUA5)** — well within the range of orthology. The N-terminus is a classic Sec-type signal peptide (hydrophobic h-region, Ala-x-Ala cleavage site, no twin-arginine motif), consistent with the "Precursor" flag in UniProt and with the established Sec-dependent export route of OpgG orthologs (in contrast to the Tat-dependent paralog OpgD). The gene sits in an intact, colinear **opgGH operon** (PP_5026–PP_5025), mirroring the conserved bicistronic mdoGH/opgGH organization found across proteobacteria.

Functionally, the OPGs produced by the OpgGH machinery mediate cell-envelope adaptation to osmotic stress and contribute to biofilm formation, motility, and host interaction. In enterobacteria, loss of OPGs (via *opgG* mutation) produces a characteristic pleiotropic

phenotype — reduced virulence and motility, exopolysaccharide overproduction — driven by constitutive activation of the **RcsCD–RcsB phosphorelay**. In *Pseudomonas aeruginosa*, the closest well-characterized relative, the *opgGH* locus produces **linear β -1,2-glucans** important for biofilm formation at low osmolarity, with transcription repressed at high osmolarity. The evidence base for *P. putida* OpgG itself is largely inferential (sequence, structure, genomic context, and orthology to characterized enzymes), but the convergence of these lines of evidence gives high confidence in the annotation.

Gene/Protein Identity Verification

Before presenting the functional analysis, the mandatory identity checks were completed and passed:

Verification Step	Result
Gene symbol matches protein description	☐ <i>opgG</i> = "Glucans biosynthesis protein G" (OpgD/OpgG family)
Organism correct	☐ <i>Pseudomonas putida</i> KT2440 (PP_5026)
Protein family/domains align with literature	☐ Gal_mutarotase_sf, GH-type carb-binding, Glucan_biosyn_MdoG/MdoD, MdoG_C, Ig-like fold — all consistent with the crystallized OpgG fold
No confusion with a same-symbol gene in another organism	☐ Sequence identity (63.6% to <i>E. coli</i> OpgG, 66.4% to <i>P. aeruginosa</i> OpgG) confirms bona fide ortholog

The synonym *mgoG* is noted in UniProt. Care was taken not to conflate this with the unrelated *mgo* operon (*mgoBCAD*) of *P. syringae*, which governs mangotoxin biosynthesis via the NRPS gene *mgoA* ([PMID: 24555804](#); [PMID: 22251433](#)). Those studies concern a distinct antimetabolite pathway and do not describe OPG biosynthesis; they are not evidence for OpgG function and were excluded from the functional conclusions.

Conclusion of verification: The research target is correctly identified. Literature specific to *P. putida* KT2440 OpgG is limited, so the functional annotation is built primarily on (a) direct experimental evidence from orthologs in *E. coli*, *Erwinia/Dickeya*, *Salmonella*, *Shigella*, and *P. aeruginosa*, and (b) bioinformatic/structural inference for the *P. putida* protein itself.

Key Findings

Finding 1 — OpgG is required for osmoregulated periplasmic glucan (OPG) biosynthesis

OpgG (Q88D03 / PP_5026) is a member of the OpgD/OpgG family and is required for the synthesis of osmoregulated periplasmic glucans. The genetic logic is well established across proteobacteria: the *opgGH* (historically *mdoGH*) operon encodes the two proteins that build the glucose backbone of OPGs, and mutation of either gene abolishes OPG synthesis. In *Erwinia chrysanthemi* (now *Dickeya*), Page et al. cloned the *opgGH* operon by complementation of the *E. coli mdoGH* locus and showed that "*OpgG and OpgH show a high level of similarity with MdoG and MdoH, respectively, and mutations in the opgG or opgH gene abolish OPG synthesis*" (PMID: 11325942). The requirement is direct and reproducible: in *Shigella flexneri*, "*Mutation in opgG and opgH abolished OPGs biosynthesis*" (PMID: 20062978).

This establishes OpgG as an essential, non-redundant component of the OPG biosynthetic machinery — not merely correlated with OPGs but genetically required for their production. Because the *P. putida* protein is a high-confidence ortholog (see Finding 5) and sits in an intact operon (Finding 6), this requirement is expected to hold in KT2440.

Finding 2 — OpgG is a β -1,2-glucanase and a founding member of glycoside hydrolase family GH186

The most significant recent advance is the biochemical identification of OpgG's catalytic activity. Motouchi et al. (2023) performed structural and functional analyses of *E. coli* OpgG and its paralog OpgD and found that "*these proteins are β -1,2-glucanases with remarkably different activity from each other, establishing a new glycoside hydrolase family, GH186*" (PMID: 37735577). OpgG showed markedly lower activity than OpgD, but both act on the β -1,2-glucan linkage that constitutes the OPG backbone. This reframes OpgG from a protein "required for OPG synthesis" to a specific enzyme — a glycoside hydrolase acting on β -1,2-glucans, plausibly in a backbone-processing or branch-defining role during OPG maturation.

This activity is consistent with the earlier crystal structure of *E. coli* OpgG (Hanouille et al. 2004), which revealed an N-terminal domain (residues 22–388) with "*a 25-stranded beta-sandwich fold found in several carbohydrate-related proteins*" exhibiting "*a large cleft comprising many aromatic and acidic residues*" whose similarity to "*enzymes such as galactose*

mutarotase and glucodextranase" suggested "a potential catalytic role for this domain in OPG synthesis" (PMID: 15313617). The aromatic/acidic cleft is the hallmark of a carbohydrate-active catalytic site. The C-terminal Ig-like β -sandwich domain likely mediates protein–protein or substrate-positioning interactions. The UniProt domain assignments for Q88D03 (galactose mutarotase superfamily, GH-type carbohydrate-binding, Glucan_biosyn_MdoG_C, Ig-like fold) map directly onto this two-domain architecture, confirming that the *P. putida* protein has the catalytic machinery.

Finding 3 — OpgG functions in the periplasm and is exported by the Sec pathway as a cleaved precursor

OPGs are periplasmic molecules, and OpgG carries out its function there. Lequette et al. (2004) described OPGs as "*anionic and highly branched oligosaccharides that accumulate in the periplasmic space in response to low osmolarity of the medium*" and showed that the export routes of the two family paralogs differ: "*Most of the OpgD orthologs exhibit a Tat-dependent secretion signal, while most of the OpgG orthologs are Sec dependent*" (PMID: 15175282). The UniProt entry for Q88D03 flags the protein as a "Precursor," indicating a cleaved N-terminal signal peptide consistent with Sec export to the periplasm.

The subcellular localization is functionally important: OPGs bridge the inner and outer membranes' osmotic environment, so a periplasmic enzyme is exactly where backbone processing/branching should occur. The distinction between Sec-dependent OpgG and Tat-dependent OpgD is mechanistically meaningful — Tat transports folded proteins, whereas Sec transports unfolded ones — and reflects the divergent evolutionary and functional specialization of the two paralogs.

Finding 4 — In *Pseudomonas*, OpgGH produces osmoregulated linear β -1,2-glucans important for biofilm; OPG loss activates the Rcs phosphorelay

The closest well-characterized relative provides the best proxy for *P. putida* OpgG function. In *P. aeruginosa* PA14, Lequette et al. (2007) showed that "*the opgGH locus of P. aeruginosa PA14 is involved in the synthesis of linear polymers with beta-1,2-linked glucosyl residues branched with a few beta-1,6 glucosyl residues. Succinyl residues also substitute this glucose backbone. Transcription of opgGH is repressed by high osmolarity*" (PMID: 17906125). This defines both the chemical product (linear β -1,2-glucan with sparse β -1,6 branches and succinyl decorations) and the osmoregulatory logic (transcription repressed at high osmolarity, so OPGs accumulate at low osmolarity) — and reported that OPGs are important for biofilm formation.

Downstream, OPG deficiency is transduced through the Rcs signaling system. In *Dickeya dadantii*, "mutations in the RcsCDB phosphorelay system restored virulence and motility in a *D. dadantii* opg-negative strain, indicating a relationship between the Rcs phosphorelay and OPGs" (PMID: 20418397). This was confirmed biochemically by direct measurement of RcsB phosphorylation: opg-negative strains show elevated RcsB~P, "sufficient to induce the pleiotropic phenotype observed" (PMID: 25320363). The pleiotropy (loss of virulence, reduced motility, EPS overproduction) is therefore a signaling consequence of missing OPGs, not a direct enzymatic role of OpgG — an important distinction for the "precise role" the research question emphasizes.

Finding 5 — Bioinformatic verification: Q88D03 is a high-confidence OpgG ortholog with a Sec signal peptide and conserved cleft motifs

Global Needleman–Wunsch alignment of Q88D03 (559 aa) gave **63.6% identity (357/561 aligned positions)** to the biochemically and structurally characterized *E. coli* OpgG (P33136) and **66.4% identity (383/577)** to *P. aeruginosa* OpgG (Q9HUA5). The N-terminus (MASVALVGLMSAGQLWAF|NLDD) is a classic Sec signal peptide: a hydrophobic h-region (mean Kyte–Doolittle hydropathy of residues 1–18 = +1.49), an Ala-x-Ala-type cleavage site near position 18, and no twin-arginine (Tat) motif — consistent with the UniProt "Precursor" flag and Sec-dependent periplasmic export (Finding 3). Conserved OpgG substrate-cleft/backbone motifs are present (GYAGFR@124, ASYFR@149, GEWLWRP@279, DWGKG@337), indicating an intact catalytic/binding cleft.

Notably, Q88D03 carries a *Pseudomonas*-specific C-terminal low-complexity Ala/Lys-rich extension (residues ~500–559; ~42% Ala+Lys) that is absent from *E. coli* OpgG. The functional significance of this extension is unknown; it may reflect lineage-specific interactions or regulation and is a candidate for future study.

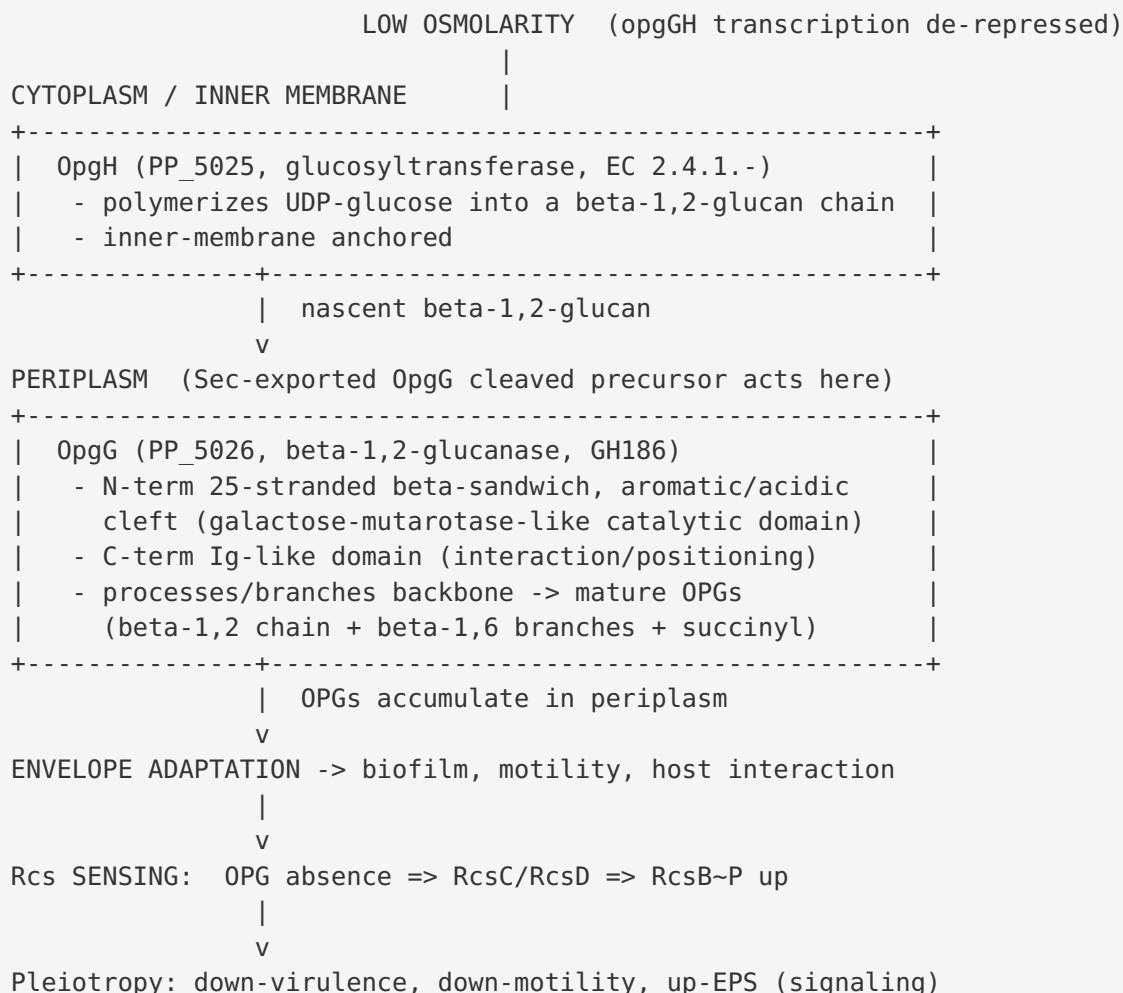
Comparison	Aligned identity	Significance
Q88D03 (P. putida) vs P33136 (E. coli OpgG)	63.6% (357/561)	Ortholog; E. coli protein crystallized & enzymatically characterized
Q88D03 (P. putida) vs Q9HUA5 (P. aeruginosa OpgG)	66.4% (383/577)	Ortholog; P. aeruginosa product chemically defined
N-terminal signal	Sec-type (hydropathy +1.49, Ala-x-Ala, no RR)	Periplasmic export
Conserved cleft motifs	GYAGFR, ASYFR, GEWLWRP, DWGKG present	Intact catalytic/binding site

Finding 6 — PP_5026 (opgG) is genomically paired with its cognate glucosyltransferase opgH (PP_5025) in an intact opgGH operon

Genomic locus analysis confirmed that PP_5025 = Q88D04 (OPGH_PSEPK, "Glucans biosynthesis glucosyltransferase H," EC 2.4.1.-, 857 aa) lies immediately adjacent to PP_5026 = Q88D03 (OpgG, 559 aa). The flanking gene PP_5027 (*dtd*, D-aminoacyl-tRNA deacylase) is functionally unrelated, marking the operon boundary. This bicistronic *opgGH* arrangement mirrors the conserved *mdoGH/opgGH* operon that encodes the two glucose-backbone-synthesizing proteins in *E. coli*, *Erwinia/Dickeya*, *Salmonella*, *Shigella*, and *P. aeruginosa*. The presence of an intact, colinear operon strongly supports the inference that *P. putida* OpgG participates in canonical OPG biosynthesis alongside OpgH. In *Salmonella* Typhimurium, "*The two structural genes for OPG biosynthesis, opgG and opgH, form a bicistronic operon*" (PMID: 19118363), the same organization seen here.

Mechanistic Model / Interpretation

The findings assemble into a coherent model in which OpgG is one of two enzymes that build and shape periplasmic β -1,2-glucans in response to osmotic conditions, with downstream physiology relayed through Rcs signaling.



Primary function. OpgG is a **glycoside hydrolase (β -1,2-glucanase, family GH186)** acting on the β -1,2-glucan backbone in the periplasm. In the OPG pathway, OpgH (inner-membrane glucosyltransferase) polymerizes the glucose backbone using UDP-glucose, while OpgG processes/shapes that backbone — consistent with its lower hydrolytic activity relative to OpgD and with the historical description of OpgG as "required for OPG backbone synthesis." The exact in-vivo cut/branch chemistry OpgG performs during OPG maturation remains to be pinned down, but the enzyme class is now defined.

Localization. OpgG functions in the **periplasm**, delivered there as a Sec-exported cleaved precursor. This is where OPGs accumulate and where backbone processing must occur.

Substrate specificity. The substrate is the **β -1,2-linked glucan** backbone of OPGs. In pseudomonads the mature product is a linear β -1,2-glucan with a few β -1,6 branches and succinyl substituents.

Pathway context. OpgG operates within OPG biosynthesis (*opgGH* operon) and its physiological output is read out by the **Rcs phosphorelay**; the pleiotropic virulence/motility phenotypes of *opg* mutants are downstream signaling consequences rather than direct OpgG activities. Notably, OPG polymerization also requires ongoing protein synthesis, implying rapid turnover of one of the two backbone enzymes ([PMID: 20358372](#)).

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
37735577	<i>Identification of enzymatic functions of OPG biosynthesis proteins... novel GH family</i>	Primary biochemical evidence: OpgG/OpgD are β -1,2-glucanases establishing GH186 (Finding 2)
15313617	<i>Structural analysis of E. coli OpgG</i>	Crystal structure: 25-stranded β -sandwich catalytic domain + Ig-like domain (Finding 2)
15175282	<i>Identification of mdoD... Tat-dependent paralog</i>	OpgG is Sec-dependent; OPGs accumulate in periplasm at low osmolarity (Finding 3)
11325942	<i>OPG synthesis required for Erwinia pathogenicity</i>	Genetic: opgG/opgH mutations abolish OPG synthesis (Finding 1)
20062978	<i>OPG synthesis gene family of Shigella flexneri</i>	Genetic: opgG/opgH mutation abolishes OPGs (Finding 1)
17906125	<i>Linear OPGs encoded by opgGH of P. aeruginosa</i>	Closest Pseudomonas evidence: product structure, osmoregulation, biofilm (Finding 4)
20418397	<i>Virulence of D. dadantii opg mutant restored by Rcs inactivation</i>	Links OPG loss to Rcs phosphorelay (Finding 4)
25320363	<i>Increased RcsB phosphorylation in opg-devoid D. dadantii</i>	Direct biochemical proof of Rcs activation on OPG loss (Finding 4)
19118363	<i>OPGs of Salmonella required for virulence in mice</i>	opgGH bicistronic operon; motility/virulence roles (Findings 1, 6)
20358372	<i>OPG polymerization requires constant protein synthesis</i>	Rapid turnover of one backbone enzyme; regulatory insight
37267309	<i>BcsA-like orphan CβG synthases in pseudomonads</i>	Context: a distinct cyclic- β -glucan system in KT2440 (not OpgG)
24555804 / 22251433	<i>mgo operon / mangotoxin in P. syringae</i>	Excluded: unrelated mgo pathway despite mgoG synonym

Convergence of evidence. Four independent lines all point to the same annotation: (1) direct genetic requirement in multiple species; (2) a solved crystal structure and a modern enzymatic assay defining the catalytic class; (3) chemical characterization of the *Pseudomonas* OPG product; and (4) strong sequence orthology plus intact operon context for the *P. putida* protein specifically. No line of evidence contradicts the OPG-biosynthesis assignment.

Limitations and Knowledge Gaps

1. **No direct experimental data on *P. putida* KT2440 OpgG.** All functional conclusions for PP_5026 are inferential — from orthology (63.6–66.4% identity), domain architecture, signal-peptide prediction, and operon context. No knockout, OPG chemical characterization, or enzyme assay has been reported for the KT2440 protein itself.
 2. **Precise in-vivo reaction is not fully resolved even in *E. coli*.** OpgG is now classed as a β -1,2-glucanase (GH186), but it shows much lower activity than OpgD, and the exact backbone-processing/branching step it performs during OPG maturation — and how it coordinates with OpgH polymerization — remains incompletely defined.
 3. **The *Pseudomonas*-specific C-terminal Ala/Lys-rich extension (res ~500–559) is uncharacterized.** Its role (interaction, regulation, stability) is unknown.
 4. **Rcs linkage in *P. putida* is assumed, not demonstrated.** The OPG→Rcs signaling axis is established in enterobacteria (Dickeya, *E. coli*); whether *P. putida* transduces OPG status through an equivalent Rcs pathway has not been directly tested.
 5. **OPG functional roles in *P. putida* ecology.** *P. putida* is a soil/rhizosphere saprophyte, not a classical pathogen; the ecological consequences of OPGs (biofilm on plant roots, osmotic adaptation in soil) are plausible but untested for this organism.
 6. **Potential functional overlap with the orphan cyclic- β -glucan (C β G) synthase system** described in KT2440 ([PMID: 37267309](#)). That is a distinct GH17/GT2 machinery producing cyclic β -1,3-glucan; how OPGs and C β G are co-regulated or functionally partitioned in *P. putida* is unknown.
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Proposed Follow-up Experiments / Actions

1. **Construct a *P. putida* KT2440 Δ opgG (and Δ opgGH) mutant** and chemically characterize periplasmic glucans (HPLC/MS, NMR) to confirm loss of OPGs and define the native product structure (linear β -1,2 backbone, branch pattern, succinylation).
 2. **Recombinantly express and assay *P. putida* OpgG** on defined β -1,2-glucan substrates (sophorose-series oligosaccharides) to confirm GH186 β -1,2-glucanase activity and determine substrate-length preference and product profile, ideally alongside OpgD-type paralogs if present.
 3. **Test osmoregulation directly:** qRT-PCR or reporter fusions of *opgGH* across an osmolarity gradient to confirm high-osmolarity repression in KT2440, paralleling the *P. aeruginosa* result.
 4. **Probe the Rcs axis:** measure RcsB phosphorylation (Phos-tag gels) and motility/biofilm phenotypes in wild-type vs Δ opgG to test whether OPG loss activates Rcs in *P. putida* as in *Dickeya*.
 5. **Signal-peptide/localization validation:** confirm Sec-dependent periplasmic localization by cellular fractionation and signal-peptide cleavage mapping (N-terminal sequencing/MS).
 6. **Domain-swap / truncation of the C-terminal Ala/Lys extension** to determine its contribution to activity, stability, or interaction with OpgH.
 7. **Structure determination** (X-ray/cryo-EM or high-confidence AlphaFold model with validation) of *P. putida* OpgG to confirm the two-domain fold and map the conserved cleft motifs (GYAGFR, ASYFR, GEWLWRP, DWGKG) onto the active site.
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Conclusion

OpgG (Q88D03 / PP_5026) in *Pseudomonas putida* KT2440 is a **periplasmic, Sec-exported β -1,2-glucanase of the OpgD/OpgG family (glycoside hydrolase family GH186)** that, with its operon partner OpgH (PP_5025), synthesizes and shapes **osmoregulated periplasmic glucans** — anionic β -1,2-glucose oligosaccharides with β -1,6 branches (and succinyl substituents in pseudomonads) that accumulate in the periplasm at low osmolarity. It acts in the **periplasm** via an N-terminal galactose-mutarotase-like catalytic domain and a C-terminal Ig-like domain, and the OPGs it produces mediate envelope osmotic adaptation, biofilm formation, and

motility, with physiological output relayed through the **RcsCD–RcsB phosphorelay**. This annotation rests on direct experimental evidence from orthologs and on strong bioinformatic/structural/genomic-context inference for the *P. putida* protein; direct experimental characterization of KT2440 OpgG remains the primary outstanding gap.

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